A NWDAF Study Employing Machine Learning Models on a Simulated 5G Network Dataset

Leonardo Azalim de Oliveira
Federal University of Juiz de Fora
Juiz de Fora, Brazil
leonardo.azalim@ice.ufjf.br

Edelberto Franco Silva
Federal University of Juiz de Fora
Juiz de Fora, Brazil
edelberto@ice.ufjf.br

Mário Antônio Ribeiro Dantas
Federal University of Juiz de Fora
Juiz de Fora, Brazil
mario.dantas@ice.ufjf.br

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***Abstract*—The 5G technology is an evolution of the mobile networks due to the adoption of a Service Based Architecture (SBA). It allows greater flexibility and the possibility of using data analysis processes in network management. The 3GPP defines the Network Data Analytics Function (NWDAF) as the network function responsible for data analysis in 5G, however, this function remains considerably unexplored in the existing literature. In order to fill this gap, this paper investigates the NWDAF using a dataset from a simulated 5G network and employing machine learning models as network protocol classifiers. The results achieved an accuracy up to approximately 75% while the average lies around 65%, providing not only reproducibility but also paving the way for future investigations in data analysis as specified by the 3GPP.**

***Index Terms*—5G, NWDAF, Network Traffic Characterization, Classification**

I. Introduction

Compared to the previous generations, one of the most important changes of the fifth-generation technology standard for cellular networks (5G) is the introduction of the Service Based Architecture (SBA) in the mobile network context. This new approach has facilitated extensive virtualization of the 5G network infrastructure, greatly enhancing flexibility in configuration and deployment of 5G-based mobile network environments. Within this context, the 3rd Generation Partnership Project (3GPP) is the most significant international collaborative body because it has specification groups especially focused on standardizing the mobile telecommunications networks. Similarly to the previous generation (4G), the network architecture of 5G [1] is specified by documents called Releases. The technical details provided in these documents serve as a comprehensive guide for implementing the associated technologies in real-world production environments.

The 5G network SBA has elements known as Network Functions (NFs). These NFs are modular components that supply specific functions and have expressive names, thereby delivering high levels of flexibility and efficiency across the network. The specification [2] introduces an interconnection between the Network Data Analytics Function (NWDAF) and other NFs aiming at establishing a cohesive and efficient network. Information exchanged between the NWDAF and other NFs through the specified Application Programming Interfaces (APIs) optimize synchronization among the 5G network services.

The NFs Analytics Logical Function (AnLF) and Model Training Logical Function (MTLF) contained in NWDAF and the Analytics Data Repository Function (ADRF), which is often used in conjunction with those, were specified with greater detail only in Release 17 [3]. Even so, a limited number of studies [4], [5] have explored this NF in the 5G Core (5GC). There is also a considerable lack of evaluation of this NF applied to typical computer network data analyses.

In order to help filling this gap, this paper uses a dataset created from a simulated 5G network as input to Machine Learning (ML) models (Decision Tree, Multilayer Perceptron, and Random Forest) that act as network protocol classifiers to explore the feasibility of this type of setup in the NWDAF. In compliance with 3GPP's Release 17, experiments on "UE Communication Information" use case were conducted. Achieving an accuracy around 75%, the best results are promising. Compared to the state-of-the-art, this work advances reproducibility and lays the foundation for future data analysis inquiries through the specified NF.

The key contributions of this study are outlined as follows:

1) Open source implementation[1] of the ML model creation functionality to be utilized by NWDAF NF;
2) Design and documentation of a testing environment containing a simulated 5G network;
3) Generation of a publicly accessible network traffic dataset captured from the testing environment; and
4) Comparison of three ML models in the context of traffic characterization of 5G User Equipments (UEs) and subsequent evaluation of the obtained results.

The remainder of this paper is structured as follows: Section II presents the related work and an outline of the state-of-the-art on the studied topic. Section III provides a comprehensive view of the proposed implementation of the ML model functionality for NWDAF, along with details about the experimental environment setup. Then Section IV delves into the results obtained from the experiments using simulated network data and includes discussions surrounding these findings. Lastly, Section V summarizes the key conclusions and outlines future research directions.

[1]https://github.com/oliveiraleo/mnc_NWDAF

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II. Related Work

This section presents the state-of-the-art regarding NWDAF and 5G architecture, highlighting the key characteristics of each work and potential future research opportunities.

The work [6] presents an investigation into the integration of data analytics in the 5G System (5GS). Requirements for this deployment are outlined, as well as the current state of standardization activities. The authors propose an enhanced SBA that integrates data analytics functionality and enables dynamic service adaptation. With a focus on predictive management of radio resources and the use of databases for radio measurements, that work assists in understanding the basics of the NWDAF and other associated NFs.

According to [7], the NWDAF plays a crucial role in conducting advanced analyses and proactive management in 5G and 6G networks. Their paper highlights that the NWDAF has a ML mechanism and integrates with the SBA. Hence, it uses the publish-subscribe pattern to retrieve information from other NFs and update them with the results of data analysis. This integration of ML-driven analyses within the NWDAF enables, for instance, anomaly detection, network load calculation, and prediction of future loads, thereby enhancing the network's ability to adapt to dynamic conditions.

Focusing on its central role in supporting service management, the NWDAF is also examined by [8]. The study explores the complexity of the NWDAF's interaction with other system modules, such as other NFs, Application Functions (AFs), or even Operations, Administration and Maintenance (OAM) functions, detailing how it collects data from multiple sources. The authors emphasize the flexibility in the deployment of the NWDAF as a Virtual Network Function (VNF), situation that allows for the presence of both single and multiple instances within the same network domain. The process of discovering and selecting the appropriate instance of the NWDAF for specific analyses is addressed, highlighting the crucial role of the Network Repository Function (NRF) in this context.

An approach to applying supervised ML for service classification is presented by [5]. The utilized methodology not only simplifies the identification of registered NFs but also enables the learning of authorized flow rules for accessing services. Therefore, the main contribution of that work lies in its approach to addressing the security of the 5GC, while offering a solution that uses ML to enhance access control.

A prominent work [4] contains a proposal for an open source implementation of the NWDAF. The proposed implementation contains both the MTLF and the AnLF modules integrated into an open source 5GC called free5GC. Although the work verified the feasibility of the NWDAF and released its source code, the step involving ML focusing on the 5G networks field remained open for future research. The MNIST dataset [9] used by the authors contains 60,000 gray scale images of numerical digits represented in matrices, so it does not include data related to network traffic.

As the 3GPP specified [2], the NWDAF is the NF responsible for collecting information from other NFs such as the Access and Mobility Management Function (AMF) and the Session Management Function (SMF). Leveraging pretrained machine learning models, it analyzes this data to provide analytical information in statistical or predictive ways.

The examination of the specifications and aforementioned work helps to identify key areas of improvement. The work [6] contributes pointing out the role of the analyses in the 5GS as future research directions, and the impact of implementing the NWDAF. It is necessary to mention the importance of the NWDAF within the complexity of 5G networks, including in the context of proactive network [7] and service management [8], and in improving the 5GC security [5]. Finally, a specified 5G use case, which, according to the information available, remained uncharted prior to this research is addressed. To achieve this, the proposal detailed in Section III was built upon [4], which provides the implementation of the NWDAF in free5GC. However, the dataset used lacks data related to Computer Networks. To address this, a 5G dataset in a well-documented simulated environment was created.

III. Proposal

Given the implementation of a NWDAF prototype integrated into a 5GC instance by [4], and the state-of-the-art overview presented in Section II, the current section introduces the proposal. As part of the validation process, the generation and utilization of metadata extracted from captured packets in a 5GC instance are presented, as well as the evaluation of utilization of ML models.

Furthermore, this section contains: an overview of an example scenario concerning a 5G network (in Section III-A); the proposed architecture based on 3GPP's specification (within Section III-B); the testing environment used for experimentation (in Section III-C); the characteristics of the dataset created in this environment (contained in Section III-D); and the conducted packet capture experiment (in Section III-E).

A. 5G Use Case Scenario

A use case scenario of the proposal involves an average user within a 5G environment. In terms of protocol classification, the elements of the evaluation environment are grouped according to an Edge-Fog-Cloud computing scheme. Through the analysis of the key elements of a 5G network and the concepts of an Edge-Fog-Cloud distributed architecture, it becomes possible to didactically separate these components. This separation is depicted in Fig. 1, where the wireless link between the UE and the Radio Access Network (RAN) is in the Edge, the 5GC is in the Fog, and other network resources (such as internet access and cloud servers) are in the Cloud. In addition to representing a hierarchical organization, which means that the UE does not access the internet without first connecting to the 5GC, this architecture allows for associating the location of elements both logically and physically.

Considering a common use case among the general public, which is a user browsing the internet using their device, it becomes clear that their first point of contact will be with the Edge components. The Fog components, even if located

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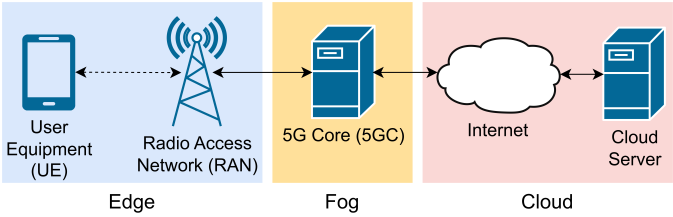

Fig. 1: Edge-Fog-Cloud architecture in the context of 5G.

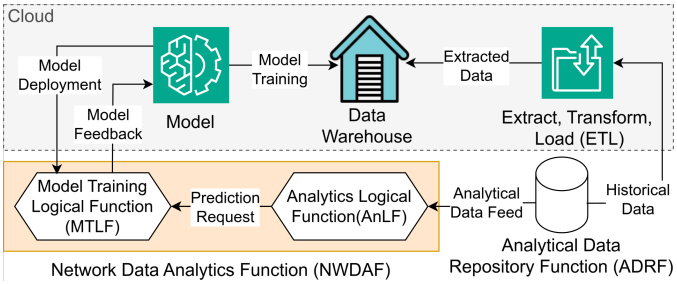

Fig. 2: Architecture based on Release 17 [2] specification.

a few hops away, can still ensure that the user's Quality of Experience (QoE) is not compromised. Similarly, in this example, the Cloud components may be located even further away from the user, but still meet the QoE requirements without significant difficulty.

In a scenario where the UE uses protocols and exchanges ping, video, and web service related data, it is possible to manage network resources based on the classification of these data flows. It is important to note that the scope of this work does not include the NFs scaling or reconfiguration based on the classified flow, but rather to serve as a foundation for future research exploring these scenarios. As previously mentioned in this paper, the chosen use case "UE Communication Information" is introduced in 3GPP's Release 17.

Following the introduction to the scenario in context and to the proposed evaluation environment, it is possible to detail the architecture with ML application within 5GC and other characteristics of the proposal.

B. Architecture Overview

The NWDAF stands out from other NFs in the 5GC for two main reasons: 1) it requires models to be retrained; and 2) it requires access to historical data. This is important because once a NF other than the NWDAF is deployed, changes on its behavior are not expected. However, a ML model is different as it is linked to the data on which it was trained. Another point to consider is that, different from other NFs that only need to know the current state of the system, the NWDAF uses recorded data. Based on the specification [2], Fig. 2 contains the proposed architecture for the data and ML pipeline. In this configuration, historical data must be collected from the ADRF and stored in a Data Lake or Data Warehouse. Data required by the AnLF for inference can come from the ADRF or other NFs. The data pipeline consists of an Extract, Transform, Load (ETL) stage, which aims to extract data from the ADRF and store it in a Data Warehouse. Then models can be trained and, finally, sent to the MTLF, which returns performance statistics in a production scenario. It is important to note that in this production scenario, both the model training stages and the storage of historical data (represented by the Data Warehouse) could be performed in a cloud environment.

C. Testing Environment Description

Based on the work by [4], [5], and [10], the previously discussed 3GPP specifications and the overview of the context presented in Section III-B, a testing environment was created.

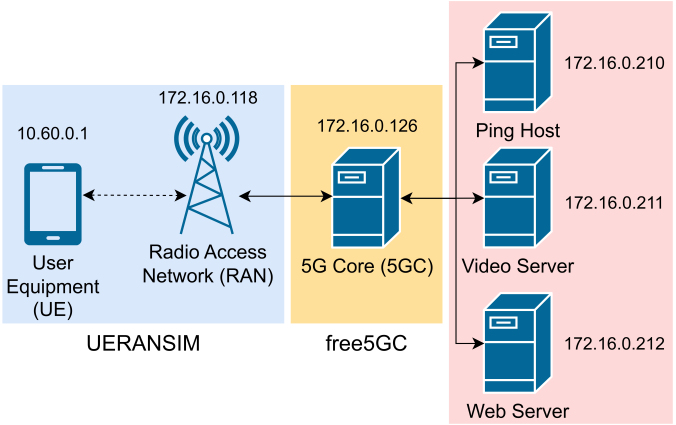

Fig. 3: Testing environment overview.

Due to a project decision, the potential benefits of using open source software, and the results available in [4] and [10], during the testing environment's design phase, open source applications were prioritized. Fig. 3 depicts this environment containing the names of the chosen applications along with the Internet Protocol (IP) addresses of each element.

The free5GC project (highlighted in yellow) implements a 5GC. It is well-documented [11] and has notable acceptance within the scientific community, with over 500 citations[2] on Google Scholar currently. The authors of free5GC recommend [12] the use of UERANSIM to perform RAN simulations. UERANSIM (colored in blue) simulates a pair of virtual UE and gNodeB (gNB) and also has considerable community acceptance, with over 330 citations[3].

In addition to these applications, three servers (highlighted in red) running ping, video, and web page services were installed. The first two servers represent situations where latency and bandwidth are Key Performance Indicators (KPIs) for user's QoE. On the other hand, compared to these services, web browsing does not have such strict requirements; therefore, each of them represents a different usage class. Server setup reproduction instructions are available in the `VMs-Setup.md` file on the GitHub repository.

[2]https://scholar.google.com/scholar?q=free5gc
[3]https://scholar.google.com/scholar?q=UERANSIM

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```
"No.","Time","Source","Destination","Protocol","Length","Info"
"1","0.000000000","10.60.0.1","172.16.0.210","ICMP","84","Echo (ping) request  id=0x0009, seq=1/256, ttl=64 (reply in 2)"
"2","0.001001426","172.16.0.210","10.60.0.1","ICMP","84","Echo (ping) reply    id=0x0009, seq=1/256, ttl=63 (request in 1)"
"3","1.002434348","10.60.0.1","172.16.0.210","ICMP","84","Echo (ping) request  id=0x0009, seq=2/512, ttl=64 (reply in 4)"
"4","1.003466096","172.16.0.210","10.60.0.1","ICMP","84","Echo (ping) reply    id=0x0009, seq=2/512, ttl=63 (request in 3)"
"5","2.004821225","10.60.0.1","172.16.0.210","ICMP","84","Echo (ping) request  id=0x0009, seq=3/768, ttl=64 (reply in 6)"
"6","2.005809609","172.16.0.210","10.60.0.1","ICMP","84","Echo (ping) reply    id=0x0009, seq=3/768, ttl=63 (request in 5)"
"7","3.006681348","10.60.0.1","172.16.0.210","ICMP","84","Echo (ping) request  id=0x0009, seq=4/1024, ttl=64 (reply in 8)"
```

Fig. 4: Sample from the ping capture training data.

D. Dataset Description

The testing environment introduced in Section III-C was employed for generating a publicly available dataset [13]. In total, the packet capture experiment was conducted six times, each limited to precisely 1000 packets. Three sets were allocated for training and validation data, while the three remaining were designated for model inference testing.

From each one of the samples, 7 features where extracted: `No.` (that represents the sequential number of the captured package), `Time` (relative time from the beginning of the capture to the moment when package was captured), `Source` (source IP address), `Destination` (destination IP address), `Protocol` (protocol tag from packet header), `Length` (packet length in bytes), and `Info` (other captured information that did not fit in any other feature). Fig. 4 depicts a screenshot displaying a sample of the data captured during the ping traffic generation. That same format was utilized in all other Comma-Separated Values (CSV) files within the dataset.

E. Explanation of the Packet Capture Experiment

The experiment consisted of creating instances of the environment's elements: servers, the 5GC, and the pair of UE and gNB, establishing connections between them, and then running scripts and commands to automate traffic generation between the UE and the three services. Throughout the test execution, the 5GC, in addition to performing its normal functionality, also conducted packet capture. The capture was executed by the TShark [14] utility on the network interface created by the GPRS Tunnelling Protocol (GTP) module installed in the free5GC instance of the environment.

To generate ping traffic, the default command of the `ping` utility available in Linux-based operating systems was utilized. The packet transmission interval was predominantly maintained at its default value of 1 second; however, packets were also transmitted in intervals of 0.20, 0.25, 0.5, 0.75, and 2 seconds. The payload size of the packet was mostly kept at its default value (60 bytes), although packets of 80 and 1400 bytes of length were also transmitted. For generating video traffic, an open source multimedia player called MPV [15] was employed. It enabled the UE to continuously playback the video file stored on the server. Lastly, web traffic was generated by executing the `wget` utility (using the file `pagelist.txt` as input parameter). As the utility was continuously executed by the UE, it was downloading the configured list of pages from the server. Following traffic capture, packets were exported in the CSV format, and these new files were then used as input for the ML models discussed in Section IV.

IV. Results and Discussion

Considering the proposal presented in Section III, this section highlights the main results achieved. First, some contributions from the initial phase of the work are explicitly stated in Section IV-A. Then, Section IV-B presents a more detailed analysis of the features present in the captured packets. Then, Section IV-C explores the ML models used, shows the statistical results along with some discussions. Finally, Section IV-D contains some considerations that are relevant to the results described in this section.

A. Initial Contributions

A validation of the work by [4] was performed. During this process, missing information was identified in the project documentation. Consequently, as part of the outcomes of this study, the supporting documentation of that work was enhanced[4]. Additionally, as a step of this process, the source code of the NWDAF module was updated to correctly work with version 3.3.0 of the free5GC project. The update was validated through an execution test in the testing environment.

B. Preliminary Exploratory Data Analysis

A preliminary exploratory analysis of the features listed in Section III-D was conducted on the dataset. It was observed that the length of the captured packets displayed the behavior depicted in Fig. 5. It displays the length frequency distribution on symmetric logarithmic scale compared to the size (in bytes) of the captured packets. It is important to mention that the web traffic had some data points (packet counts) trimmed: 4 packets of 93, 247, 454, 581 and 892 bytes; 5 packets of 190 bytes; 6 packets of 189 bytes; and 8 packets of 199, 201 and 223 bytes totaling the 55 packets represented on Fig. 5.

Notably, similar capture patterns are observed between inference and training sets for the same service. For example, ping packets predominantly consist of 84-byte packets, while video and web services exhibit a distribution with peaks at 69 bytes and 1400 bytes. The main distinction lies in the packet sizes, with video traffic concentrated at 52 and 1400 bytes, and web traffic more evenly distributed across measured values.

Probably, this behavior occurred due to the nature of the traffic of the services. On one hand, the video file has tens of megabytes in length and is transmitted as a continuous stream of data. On the other hand, web browsing involves the transmission of objects of different sizes that are loaded as the script downloads the different pages. Finally, the default size of the ping from the Linux operating system used to execute the virtual UE was 60 bytes of payload plus the packet header.

From another perspective, Fig. 6 presents the occurrence of protocol tags given by TShark in each one of the three captures conducted to create the inference set. It is possible to observe that ping exhibits distinct protocol patterns, while video and web services primarily utilize Transmission Control Protocol (TCP) and Hypertext Transfer Protocol (HTTP).

[4]https://github.com/net-ty/mnc_NWDAF/pull/8

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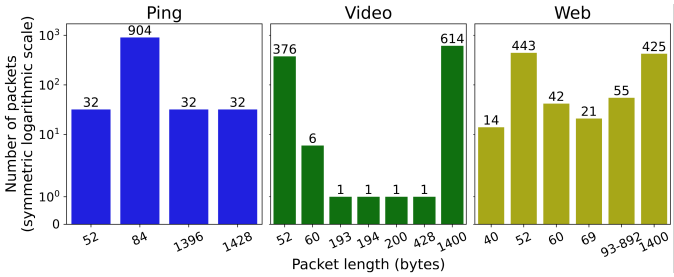

Fig. 5: Packet length distribution of the dataset's training set.

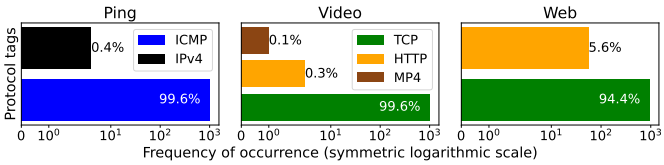

Fig. 6: Packet protocol tags in the dataset's inference set.

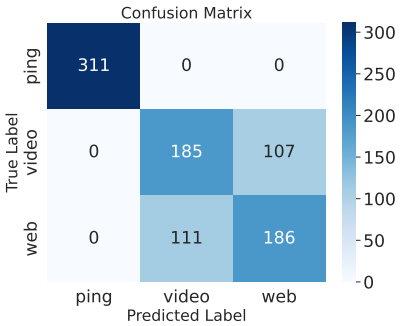

Fig. 7: Random Forest confusion matrix of classes.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.758	0.758	0.758	0.758
Multilayer Perceptron	0.659	0.655	0.755	0.751
Decision Tree	0.743	0.740	0.751	0.755

TABLE II: Model performance metrics.

Two main reasons can be identified as responsible for these behaviors: 1) the way the services operate (or more specifically, which protocol they use), as TCP was likely employed for data transmission itself, while HTTP was responsible for controlling and commanding the requests made to the servers; and 2) the manner in which the video and web services were configured, through the use of servers with similar operation characteristics, which possibly contributed to influencing the results of the ML models discussed in the following Section IV-C.

C. Usage of Machine Learning Models

A possible additional improvement specified [3] for the NWDAF is the integration and utilization of ML. In this context, three distinct models were employed: Decision Tree [16], Multilayer Perceptron [17], and Random Forest [18].

The Decision Tree was chosen due to its capability to handle mixed data types, large and complex datasets, and ease of visualization, while being resistant to missing values and requiring minimal preprocessing. Multilayer Perceptron exhibits excellent performance, accuracy, and ease of application in both linear and non-linear problems. Random Forest offers robustness to overfitting, implicit feature selection, and scalability in large datasets.

To assess the models' efficacy, a test set comprising a 30% portion of the training dataset was employed to calculate the performance metrics. These metrics, including their optimal outcomes, are represented in Table II. It is observed that both Decision Tree and Random Forest achieved accuracy and precision exceeding 70%. However, in the subsequent metrics, Random Forest showed a slightly better performance, reaching 75.8% in precision, recall, and F1-score, respectively.

Model	Protocol	Length	Duration
Random Forest	0.622	0.287	0.091
Decision Tree	0.962	0.038	0

TABLE I: Feature importance for Decision Tree and Random Forest models.

A confusion matrix is included in Fig. 7. Through its examination, it can be seen that the Random Forest correctly predicted 100% of the ping class, 63.5% of video, and 62.6% of web. It should be noted that, due to the measures adopted in the code to enhance result reproducibility, both the confusion matrix and the best performance results tend to converge towards those presented in this section.

Aiming for identifying which attributes were most relevant for the models, the importance of each attribute was extracted. As seen in Table I, it is evident that the attribute `protocol` plays a crucial role in identifying particular transfers. This distribution of importance, combined with the results of the analyses in Subsection IV-B, may help explain the results of the matrix in Fig. 7, as it was easier to differentiate the ping class from others based on the protocol used.

Furthermore, as part of the results directly related to the usage of ML models, class inference tests were conducted with sets present in the dataset but not incorporated into the training phase. Table III presents detailed results of this inference process. Each packet was classified by the models and the sum of this classification was used to give the final results that are highlighted in bold. The file names are related to the experiments in which they were generated.

Notably, a 10-iteration differential evolution process [19] was performed to fine-tune model parameters. The unbalanced training data, a consequence of the subset creation method, may have impacted the results.

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	Sample	1ping-capture-intervals			2video-capture			3web-capture		
Random Forest	Classes	**ping**	video	web	ping	**video**	web	ping	video	**web**
	Inference	**999**	–	–	–	**688**	311	–	424	**575**
Multilayer Perceptron	Classes	**ping**	video	web	ping	**video**	web	ping	video	**web**
	Inference	**996**	3	–	3	**684**	312	57	424	**518**
Decision Tree	Classes	**ping**	video	web	ping	**video**	web	ping	video	**web**
	Inference	**999**	–	–	–	**685**	314	–	424	**575**

TABLE III: Inference results.

D. Final Comments

The results from Section IV-C were obtained by excluding the source and destination IP addresses from the models. This action was done because as the models learn the services' static IPs, the accuracy reaches 100%. It was possible to confirm that Random Forest and Decision Tree attribute 49.3% of the importance to the IPs. This discovery holds relevance as it presents potential applicability in analogous environments.

Considering the context and the findings of this research, it would be desirable to offer real-time learning capabilities. As per 3GPP specifications, this would involve the ability to generate and update the model with real-time data. The proposed approach to achieve this enhancement from the current state of the presented solution comprises: 1) monitoring of one or more KPIs, triggering model updates upon reaching a configured threshold; or 2) operator-initiated updates via the Control Plane (CP). Given the substantial quantity of data that would be produced, a storage-based technology solution might be required in conjunction with a Data Warehouse system.

V. Conclusion

This paper investigates the applicability of machine learning models in the NWDAF for traffic classification on a 5G simulated environment. Building upon the work of [4] this study contributes to the Computer Networks field by evaluating traffic classification using ML models over a 5G network dataset while providing documentation and reproducibility methods. The results demonstrate promising accuracy values of up to 75% using simple data analysis models. The designed testing environment is adaptable to other 5G-related experiments, enabling the creation of additional 5G datasets. The generated dataset is reusable for other research groups to analyze 5G simulated traffic, facilitating further research and development.

As future work, it is envisioned the complete integration of the ML module with the NWDAF module that connects to the official free5GC while complying with the specification [2]. It is also planned the exploration of other use cases within those specified by 3GPP in [3] as well as expanding the services available in the testing environment. Finally, developing an open source experimentation framework which not only allows the reproduction of results but also serves as a foundation for future validations regarding the NWDAF is planned.

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References

[1] A. Sultan, "5g system overview," Aug 2022. [Online]. Available: https://www.3gpp.org/technologies/5g-system-overview

[2] 3GPP, "3GPP TS 29.520 version 17.11.0 Release 17," 3rd Generation Partnership Project, Tech. Rep., 07 2023.

[3] ——, "3GPP TS 23.288 version 17.4.0 Release 17," 3rd Generation Partnership Project, Tech. Rep., 05 2022.

[4] T. Kim, J. Kim, H. Ko, S. Seo, Y. Jcon, H. Jeong, S. Lee, and S. Pack, "An implementation study of network data analytic function in 5g," in *2022 IEEE International Conference on Consumer Electronics (ICCE)*. IEEE, 01 2022, p. 1–3.

[5] R. Pell, S. Moschoyiannis, and M. Shojafar, *Machine Learning Driven Secure Access Control for 5g Core Networks*. SSRN, 2023.

[6] E. Pateromichelakis, F. Moggio, C. Mannweiler, P. Arnold, M. Shariat, M. Einhaus, Q. Wei, O. Bulakci, and A. De Domenico, "End-to-end data analytics framework for 5g architecture," *IEEE Access*, vol. 7, p. 40295–40312, 2019.

[7] E. Coronado, R. Behravesh, T. Subramanya, A. Fernàndez-Fernàndez, M. S. Siddiqui, X. Costa-Pérez, and R. Riggio, "Zero touch management: A survey of network automation solutions for 5g and 6g networks," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2535–2578, 2022.

[8] Q. Duan, "Intelligent and Autonomous Management in Cloud-Native Future Networks," *Future Internet*, vol. 13, no. 2, p. 42, 02 2021.

[9] L. Deng, "The mnist database of handwritten digit images for machine learning research," *IEEE Signal Processing Magazine*, vol. 29, no. 6, pp. 141–142, 2012.

[10] L. A. de Oliveira and E. F. Silva, "Evaluation of eap usage for authenticating eduroam users in 5g networks," 2024.

[11] free5GC, "User guide," 2024. [Online]. Available: https://free5gc.org/guide/

[12] ——, "Tested hardware," 2024. [Online]. Available: https://free5gc.org/guide/hardware/

[13] L. A. de Oliveira, "Nwdaf packet dataset folder," 2024. [Online]. Available: https://github.com/oliveiraleo/mnc_NWDAF/tree/mnc_Public-5G/ML_test_code/dataset

[14] "tshark - the wireshark network analyzer 3.2.2," 2024. [Online]. Available: https://www.wireshark.org/docs/man-pages/tshark.html

[15] "Mpv," 2023. [Online]. Available: https://mpv.io/

[16] S. R. Safavian and D. Landgrebe, "A survey of decision tree classifier methodology," *IEEE transactions on systems, man, and cybernetics*, vol. 21, no. 3, pp. 660–674, 1991.

[17] S. K. Pal and S. Mitra, "Multilayer perceptron, fuzzy sets, classifiaction," 1992.

[18] L. Breiman, "Bagging predictors," *Machine learning*, vol. 24, no. 2, pp. 123–140, 1996.

[19] R. Storn and K. Price, "Differential Evolution – A Simple and Efficient Heuristic for global Optimization over Continuous Spaces," *Journal of Global Optimization*, vol. 11, no. 4, p. 341–359, 1997.

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